

NAKHON PHANOM, Thailand

WMO: 483570

Lat: 17.410N

Lon: 104.776E

Elev: 486

StdP: 14.44

Time Zone: 7.00 (E07)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	56.5	59.1	44.9	44.7	64.8	48.0	50.5	65.1	11.0	74.6	9.7	75.0	0.3	90	0.288

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
4	16.8	98.4	78.5	96.2	78.1	94.4	78.0	82.3	91.2	81.4	89.8	80.8	88.9	2.5	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
80.1	159.0	86.4	79.3	154.9	85.2	78.9	152.5	84.4	46.6	91.4	45.6	90.0	44.9	89.2	89.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.0	7.6	6.8	50.5	101.5	3.2	2.3	48.2	103.1	46.4	104.5	44.6	105.8	42.3	107.5		
(5)			48.1	84.5	3.1	2.1	45.9	86.0	44.1	87.2	42.4	88.4	40.2	89.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	80.4	73.3	77.2	82.1	85.3	84.7	83.4	82.0	81.9	82.1	81.3	78.1	73.7	(6)
(7)		DBStd	5.45	4.66	5.58	5.37	4.24	3.30	2.77	2.87	2.47	2.44	2.71	3.78	4.77	(7)
(8)		HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)
(9)		HDD65	8	3	2	0	0	0	0	0	0	0	0	0	3	(9)
(10)		CDD50	11106	721	761	996	1058	1075	1003	993	989	962	970	843	735	(10)
(11)		CDD65	5639	259	343	531	608	610	553	528	524	512	505	393	273	(11)
(12)		CDH74	58892	2143	3260	5946	7548	7203	6183	5369	5228	5164	5069	3533	2246	(12)
(13)		CDH80	23967	710	1440	2939	3924	3273	2415	1766	1636	1796	1990	1347	731	(13)
(14)	Wind	WSAvg	1.5	2.1	2.1	1.9	1.5	1.2	0.9	0.8	0.9	0.9	1.4	1.9	2.2	(14)
(15)	Precipitation	PrecAvg	95.4	0.1	0.5	2.0	3.5	10.2	22.2	17.0	26.0	11.7	2.4	0.2	0.1	(15)
(16)		PrecMax	111.7	0.4	1.9	4.6	8.0	17.4	40.7	30.4	48.5	23.1	11.5	1.8	1.1	(16)
(17)		PrecMin	71.7	0.0	0.0	0.2	0.4	6.5	11.3	10.7	10.8	3.8	0.3	0.0	0.0	(17)
(18)		PrecStd	13.7	0.1	0.6	1.5	2.2	2.7	8.9	4.9	10.4	5.2	2.7	0.4	0.3	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	92.1	96.8	101.1	102.1	99.1	95.2	94.0	93.3	93.1	93.1	93.2	91.5	(19)
(20)			MCWB	72.5	73.4	76.0	79.6	80.7	80.8	80.6	80.7	79.2	77.3	76.4	74.1	(20)
(21)		2%	DB	89.0	94.1	97.6	99.1	96.3	93.2	91.8	91.0	91.6	91.5	90.9	88.8	(21)
(22)			MCWB	71.7	74.1	76.6	79.3	80.1	80.3	80.2	80.2	78.8	76.9	75.3	73.1	(22)
(23)		5%	DB	86.7	91.5	95.2	96.7	94.1	91.4	89.8	89.3	90.1	90.3	89.2	86.6	(23)
(24)			MCWB	70.8	73.8	76.5	78.8	79.7	80.0	80.0	79.8	78.5	76.5	74.6	71.8	(24)
(25)		10%	DB	84.3	88.6	92.6	94.2	91.8	89.6	87.9	87.5	88.5	88.8	87.2	84.3	(25)
(26)			MCWB	69.7	72.8	75.9	78.3	79.4	79.8	79.6	79.4	78.2	75.9	73.5	70.4	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	76.4	78.8	80.4	83.7	85.1	82.5	81.9	81.9	81.5	80.4	78.5	76.8	(27)
(28)			MCD B	86.6	90.5	94.1	96.1	93.9	91.0	89.7	89.2	88.9	88.6	88.0	86.6	(28)
(29)		2%	WB	74.3	77.0	78.9	81.7	82.5	81.4	81.0	81.0	80.7	79.1	77.5	74.6	(29)
(30)			MCD B	83.8	88.7	92.2	93.4	91.2	89.5	88.3	87.9	87.9	87.2	86.5	85.1	(30)
(31)		5%	WB	72.5	75.6	77.9	80.4	81.0	80.8	80.5	80.5	80.0	78.2	76.4	73.3	(31)
(32)			MCD B	82.8	86.6	90.6	92.0	89.9	88.5	87.4	87.0	86.7	86.0	85.1	84.0	(32)
(33)		10%	WB	71.0	74.3	77.0	79.4	80.2	80.2	79.9	79.8	79.3	77.5	75.3	71.8	(33)
(34)			MCD B	81.9	84.8	89.0	90.6	88.7	87.3	86.1	85.7	85.6	84.8	83.7	82.3	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	20.8	20.6	18.9	16.8	13.7	11.4	10.0	10.2	11.6	14.7	17.6	19.8	(35)
(36)			MCD BR	22.6	23.3	21.0	19.1	16.4	13.8	13.2	13.2	14.5	16.1	18.5	20.6	(36)
(37)			MCWBR	8.1	7.2	6.1	5.8	5.1	4.3	4.4	4.4	4.6	4.8	5.8	7.2	(37)
(38)		5% WB	MCD BR	20.3	20.2	19.2	17.1	14.4	12.5	11.8	12.1	12.6	14.1	16.2	19.1	(38)
(39)			MCWBR	8.5	8.0	6.6	6.4	5.6	4.6	4.5	4.6	4.7	4.9	5.7	7.5	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.452	0.559	0.663	0.633	0.512	0.475	0.462	0.474	0.505	0.512	0.454	0.455	(40)	
(41)		taud	2.173	1.894	1.694	1.800	2.153	2.281	2.312	2.266	2.170	2.115	2.250	2.185	(41)	
(42)		Ebn at Noon	262	239	220	228	253	261	264	263	256	249	260	257	(42)	
(43)		Edh at Noon	45	62	78	71	49	43	42	44	48	49	42	43	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1483	1586	1581	1690	1614	1477	1365	1364	1433	1500	1477	1398	(44)	
(45)		RadStd	88	115	66	91	127	141	167	117	128	130	68	55	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (10 neighbors)		+0.54	N/A	N/A	N/A	+0.36	+0.27	N/A	N/A	N/A	N/A

Nomenclature: See separate page